Exp. No: 02

STUDY OF DIFFERENT TYPES OF NETWORK CABLES

AIM:

To study the different types of network cables and prepare straight through and crossover ethernet cables.

Cable type	Category	Maximum Data Transmission.	Advantages / Disadvantages.	Application.
UTP	Category 3 Category 5 Category 5e	10 bps Upto 100 Mbps 1 Gbps.	* Cheaper in cost * Easy to install as they have a smaller overall diameter. * More prone to EMI	* 10 Base-T Ethernet * Fast Ethernet, Gigabit Ethernet
STP	Category 6 6a	10 Gbps.	* Shielded * Faster than UTP	* Widely used in data centres.
SSTP	Category 7	10 Gbps	* Expensive * Greater installation.	* Gigabit Ethernet
Coaxial Cable.	RG-6 RG-59 RG-11	10-100 Mbps	* high bandwidth * Limited distance * Cost * Size is bulky.	* Speed of signal is 500m. * High speed internet connection.
Fibre Optics Cable	Single mode Multimode	100 Gbps.	* high speed. * high security. * Expensive. * Requires skilled installers.	* Maximum distance of fibre optics cable is around 100 metres.

PROCEDURE:

* Collect the Ethernet cable, RJ45 connectors and crimping tool.
* Strip about 1.5 cm of the cable jacket from both ends.
* Arrange the wires in T568A or T568B color sequence.
* Trim the wires and insert them into the RJ45 connector.
* Crimp the connector using the crimping tool.
* Repeat for the other end.
* Test the cable by connecting two devices.

RESULT:

Thus, the straight-through and cross over Ethernet cables were successfully prepared and tested, and proper network connectivity was verified.